Work proposal for
Medicinal and Synthetic Chemistry RFP

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1. Executive Summary:

This RFP is submitted by Aurigene Pharmaceutical Service Limited ("APSL") to Johnson & Johnson (JNJ), an American multinational, pharmaceutical, and medical technologies corporation, for Medicinal and Synthetic Chemistry services.

APSL has the capability, capacity and experience to deliver projects in all type of services as described in JNJ's RFP. Our response is designed to reflect our teams' chemistry capabilities, experience and strength in meeting the proposed deliverables with high quality and fast turnaround time.

We are governed by a strong corporate framework, customer centric systems and processes which meet global standards of safety, quality, and data integrity. APSL does not carry out any in-house drug discovery projects and has robust mechanism in place to guard customer confidentiality.

We thank JNJ for their time and consideration and look forward to further discussions on our proposal.

2. Introduction to Aurigene Pharmaceutical Services Limited

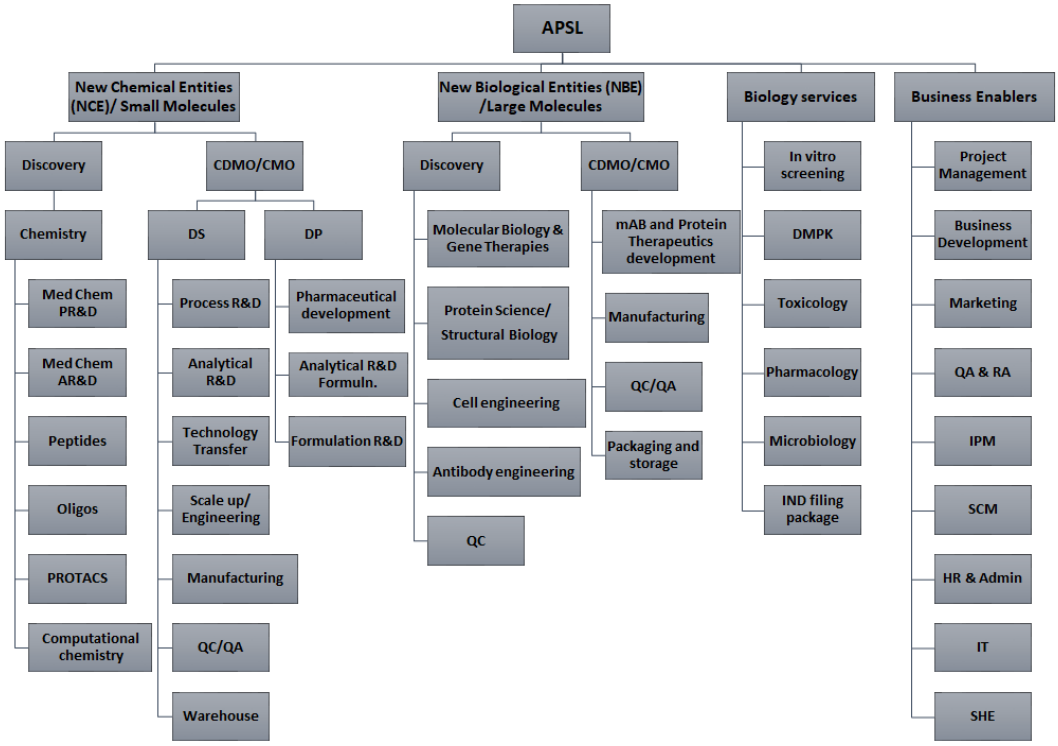
APSL is a leading contract research, development and manufacturing organization (CDMO) providing end-to-end solutions. With a strong legacy we are ideally positioned to serve global pharma and specialty companies worldwide.

Our end-to-end integrated services offerings range from chemistry, biology, custom development and manufacturing of clinical compounds to enabling IND/NDA fillings and commercial contract manufacturing. Our discovery services are offered from our world-class R&D centres situated at Bangalore & Hyderabad in India. The discovery services are spearheaded by ~880 scientists working across the drug discovery value chain with more than 100 of them having Ph.D.'s with post-doctoral experience from western research labs. Our development capabilities are supported by multiple scale-up and commercial manufacturing facilities in India, UK, Mexico and the United States.

Our endeavour is to help accelerate the journey from laboratory to market by providing access to our capabilities, knowledge and experience. APSL is committed to continuous investment in acquiring world class talent and innovative technologies, *vis-a-vis* expansion of capacity in all areas of service.

While accelerating the R&D programs of our clients, we keep sustainability, quality, environment, health and safety (EHS) at the forefront of everything we do.

2.1 Organization structure



Governance Mechanism:

Strong governance mechanism for effective partnership with client: program governance has multi-level oversight to ensure successful execution

Program Execution Team
Reviews progress at program team level
Client and Aurigene: Program lead, Project manager, SPOC, scientific team — Weekly/Biweekly

Functional Review Team
Operationalizes program level decisions
Client and Aurigene: SRC, Project Execution Team — 1-2 months

Governance Council
Drives strategic collaboration goals
Client and Aurigene: JSC, SRC, Project Execution Team — Quarterly or half-yearly

SRC: Senior Research Council
JSC: Joint Steering Committee

The services quoted in this proposal will be executed by the New Chemical Entities (NCE) team at APSL.

2.2 Site of execution

The proposed project services in the proposal will be executed from our state-of-the-art facility located in Bangalore and Hyderabad, India.

The scientific team that will be involved in the project comprises of scientific experts having experience in diverse fields of small molecule discovery. The scientists have post docs from western universities and industrial experiences in relevant area of research. We have scientists with mixed experience and expertise who are masters graduates with 5-15 years of experiences; PhDs with 5-10 years of industrial experience in the relevant field and Post docs with 5-10 years of industrial experience in the relevant field. We maintain a PhD to Masters ratio of 1:9.

3. Scope of services

Below are the chemistry activities that will be offered under non-GMP conditions as suggested in the RFP

- Synthetic chemistry services
- Route scouting and development
- Library synthesis
- Access to FFS synthesis
- Building block synthesis/acquisition
- Access to scale up synthesis and purification up to 500-gram scale
- Chiral separation services
- Compound management services

4. Key Project Assumptions:

- Information sharing: APSL expects to establish an effective mechanism of information sharing between customer and itself which will include, when applicable, knowledge on prior art, technologies, processes, and analysis methods.
- Procurement: APSL will identify and acquire all goods, services and/or works from external sources ensuring quality, quantity, turn-around time. Vendor qualification, raw material and consumables testing is not considered in this proposal. All raw material and consumables will be released based on Certificate of Analysis (CoA) from the supplier.
- Documentation: APSL will generate all the documentation required to execute the project as per internal guidelines
- Quality Control and Assurance: All activities in this proposal are non-GMP.
- Outsourcing: APSL will engage appropriate third-party (ies) for specifically selected activities, tests and analyses currently not available in APSL, upon client's consent (e.g. large scale chiral purification, trace metal analysis, XRD services, etc).

5. Business Proposal

5.1. FTE Pricing model

In this section, APSL proposes an FTE arrangement with flexibility in deploying FTEs as per the needs of the project. Based on the progress and requirement in the project, JNJ will have an option

to extend the period of collaboration or increase/ decrease the number of FTEs as desired, with at least 30 days advance notice to APSL.

a) FTEs will be charged at annual rate as per the table below

Business service	FTEs Range	FTE price/ annum (USD)
Chemistry FTE	up to 10	$70,000.00
	10-25	$69,000.00
	26-50	$68,000.00
	51-100	$67,000.00
	101-150	$66,000.00
	151-200	$65,000.00

- Each project shall have a team composition of billable bench scientists in a ratio of 1:5 [Phd/Post Doc (1): MS (5)], distinguishing between PhD/Postdoc professionals with a minimum of 2-3 years of relevant industrial experience and MS professionals with 8-10 years of pertinent industrial expertise. The team will be supervised by a seasoned PhD/Postdoc leader boasting 8-10 years of experience. It's important to note that such leaders will not be billed to JNJ as per our policy.

b) Deployment of additional FTEs apart from originally agreed FTEs will be allotted upon mutual agreement between both JNJ and APSL, based on the project requirement and progress in each phase of the program and upon a formal approval from JNJ.

c) All FTE teams will be led by a technical project leader with relevant industrial R&D experience, excellent in English communication, strong scientific skills and troubleshooting capability.

d) Additionally, APSL will deploy a non-billable project manager to co-ordinate operational aspects of the project.

e) APSL will also be responsible for providing sufficient other personnel as necessary to perform all administrative and other non-scientific activities to support the services.

5.2. Terms and conditions: FTE Model

5.2.1. FTE Price Validity and Invoicing

- The pricing is valid for period of one (01) year from the start date as suggested by JNJ.
- FTE billing will be done, for the actual number of FTEs agreed between JNJ and APSL during the said period, on monthly basis.

5.2.2. Consumables cost:

a) FTE Pricing include cost of standard reagents, chemicals and consumables for chemistry services, captured in Annexure 1.

b) FTE Pricing does not include cost of project specific reagents, chemicals, speciality consumables, speciality solvents for chemistry services. Aurigene will share the list of all raw materials and consumables for JNJ ' s confirmation ahead of raising the invoice.
c) The invoice for all consumables will be raised at the end of each month and same shall be paid by JNJ on a monthly basis.

5.2.3. Chiral separation

a) The FTE cost is inclusive of chiral separation up to 25 mg scale of final output /product and will not be billed to JNJ.
b) Any outsourced study cost will be borne by JNJ at actuals.

5.2.4. Shipping costs:

a) FTE pricing includes cost of once-a-week normal shipment to JNJ. Cost of dry ice shipment is not included in the above pricing. Any additional shipments in a week and dry ice shipments with or without data logger will be billed to JNJ at actuals.
b) Shipping of all samples will be done by certified express carriers in accordance with DAP Incoterm 2010, Aurigene will ensure all essential documentation for the shipment
c) Duty & clearance charges for chemicals/ test items shipped by the JNJ to APSL specific to project, if any, will be charged at actuals as pass through costs.

5.2.5. Payment Terms:

a) Aurigene shall invoice JNJ for the agreed upon FTEs during the said period on monthly basis, on the 5th of the month for the last month. For example, January invoice shall be sent in PDF by February 5th, and a hard copy sent by courier without any delay.
b) The invoice for consumables and shipping costs at actuals will be raised on monthly basis, on the 5th of the month for the last month. Aurigene shall seek pre-approval from JNJ for such reimbursable consumables.
c) All pass-through cost will have an administrative fee of 10%.
d) All Payments will be due within ninety (90) days/ as per MSA from the date of invoice.
e) Payments for reimbursable will be invoiced on a monthly basis.

6. Communication, Reports and Update Schedule

The formats of all reports will be mutually decided by the Project leaders according to the internal needs of JNJ.
Weekly short reporting will be established to ensure a steady flow of information to JNJ.

The full report and the KPIs will be delivered one week after end of month.

- Regular web/ Telecon (TCs) with project leaders and responsible scientists from JNJ will be scheduled routinely as mutually agreed between JNJ and Aurigene scientific teams, co-ordinated by a Project Manager.
- Reporting and TCs will be scheduled in a way that the reports are available 3 days in advance of the TC.

- At any time, questions and potential issues will be addressed by email and by direct phone calls.
- JNJ will provide a share point for data storage and transfer. Aurigene can also support in generation of dedicated share point for this program on need basis, if suggested by JNJ.
- JNJ will provide 2-3 weeks advance information to Aurigene about forthcoming projects to help Aurigene complete raw material procurement and other project scheduling related activities.

7. Documentation

Experimental information will be entered into the notebooks in full accordance with the JNJ notebook policy and SOP or as mutually agreed between JNJ and APSL.
JNJ will provide APSL the SOP 'Instructions for use of laboratory notebooks' and APSL guarantees that all involved FTEs observe the guidelines of the SOP.

8. Intellectual Property

APSL will ensure that Intellectual Property generated during the term of this project shall be transferred to JNJ. JNJ will exclusively own all data generated for the project (s).

9. Other

JNJ may terminate the project with a notice period of thirty (30) days in case JNJ considers that the project within which JNJ entrusted APSL with the performance of the services is not of interest any more for any reasons or that following preliminary results obtained during the performance of the services holds no longer any interest to have the services performed by APSL.

In case of termination for any cause whatsoever, APSL shall (i) return to JNJ all data, documents and remaining quantities of materials, samples, and product supplied, (ii) provide JNJ with a report concerning the services as well as the results already obtained at the effective date of termination. JNJ shall pay to APSL the sums due in consideration of the services actually performed and for reimbursable expenses reasonably incurred up to the effective date of termination, including unavoidable costs for product-specific reagents already ordered or otherwise resources committed at the time of termination.

10. Validity

This proposal is valid for 90 days from the date of submission.
Should there be any questions or clarifications needed, please feel free to contact **Jayachandran Ezhuthachan.**

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11. Authorization

Each party's signature further constitutes acceptance of the price, schedules, payment terms and other terms and conditions set out or referred to in this quotation.

Johnson & Johnson	**Aurigene Pharmaceutical Services Ltd.**
By _____	By _____
Name _____	Name _____
Title _____	Title _____
Date _____	Date _____

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12. Anexure-1

Standard Laboratory Chemicals

Chemical	Specification	CAS No
Alumina, Neutral	Aluminium oxide 90, 70-230 mesh ASTM	1344-28-1
Ammonia solution, 25%	28.0-30.0% as NH3	7664-41-7
Ammonium chloride	≥99.5%	12125-02-9
ammonium hydroxide		1336-21-6
Anti-bumping granules		
Calcium chloride	technical grade, −4-+30 mesh	10043-52-4
Celite 545		61790-53-2
Activated Charcoal	decolorizing agent	7440-44-0
Citric acid	≥99%	77-92-9 99
Cuprous chloride	97%	7758-89-6
Drierite, 8 mesh, no indicator	without indicator, 8 mesh	7778-18-9
Hydrobromic acid		10035-10-6
Hydrochloric acid (sg 1.18)	37%	7647-01-0
lithium hydroxide	≥98%	1310-66-3
Magnesium sulfate	anhydrous, reagent grade, ≥97%	7487-88-9
Molecular sieves		70955-01-0
Ninhydrin spray		485-47-2
Nitric acid	70%	7697-37-2
Phosphomolybdic acid	95%	51429-74-4
Phosphoric acid		7664-38-2
Phosphorus pentoxide	≥98.0%, powder	1314-56-3
Potassium acetate		127-08-2
Potassium bicarbonate		298-14-6
Potassium carbonate	99%	584-08-7 99
Potassium fluoride	≥99.0%	7789-23-3
Potassium hydroxide	≥90%, flakes	1310-58-3
potassium phosphate tribasic		7778-53-2
potassium phosphate dibasic		7758-11-04
potassium phosphate monobasic		7778-77-0
Sand	−50-70 mesh particle size	14808-60-7
Sieves, 3 angstrom	pellets, 1.6 mm	308080-99-1
Sieves, 4 angstrom powder	powder, activated, 5 µm	70955-01-0
Sieves, 4 angstrom	pellets, 1.6 mm	70955-01-0
Silica gel with moisture indicator (blue gel), dessicant ~1-3mm		112926-00-8
Silicone oil, high temp. (230 C)	Silicone oil, high temp. (230 C)	63148-62-9
Sodium acetate		127-09-3
Sodium bicarbonate	≥99.5%, crystalline	144-55-8

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Chemical	Specification	CAS No
Sodium carbonate	≥99.0%	497-19-8
Sodium chloride (AnalaR)	≥98%, 80 mesh particle size	7647-14-5
Sodium hydride, 60% in oil	60 % dispersion in mineral oil	7646-69-7
Sodium hydroxide	97%, beads	1310-73-2
Sodium sulfate	≥99.0%, anhydrous, granular	7757-82-6
Sodium sulfite		7757-83-7
Sodium thiosulphate.5H2O	Pentahydrate, 99.5%	10102-17-7
Sodium iodide	≥99%	7681-82-5
Solka floc		
Sulfuric acid	95-98%	7664-93-9
Acetic acid	glacial, 99.8%	64-19-7
Acetone	≥99%	67-64-1
Acetonitrile (LC/MS)	LC-MS grade, ≥99.9%	75-05-8
Acetonitrile	99%	75-05-8
Chloroform	≥99%	67-66-3
Cyclohexane		110-82-7
1,2-dichloroethane	≥99.0%	107-06-2
Dichloromethane	≥99.5%,	75-09-2
Diethyl ether, AR	≥99.0%	60-29-7
Diisopropyl ether		108-20-3
N',N-dimethylacetamide		127-19-5
N,N-Dimethylformamide	≥99.8%	68-12-2
Dimethylsulfoxide	≥99.9%	67-68-5
1,4-dioxane	≥99.0%	123-91-1
Ethanol	reagent grade, denatured	64-17-5
Ethyl acetate	≥99.5%,	141-78-6
n-Heptane	99%	142-82-5
Hexanes		110-54-3
Methylcyclopentane		96-37-7
Isopropyl acetate		108-21-4
Methanol	≥99.5%,	67-56-1
Methanol (for LCMS)	LC-MS grade, 99.9%	67-56-1
Methyl tert-butyl ether		1634-04-4
2-propanol	≥99%	67-63-5
Tetrahydrofuran	≥99.0%, Inhibitor-free	109-99-9
Toluene	≥99.5%	108-88-3
Anisole	99%	100-66-3
Triethylamine	≥99%	121-44-8
Pyridine	≥99%	110-86-1